Section A: Multiple Choice Questions

Thermochemistry and Energy PAST SAC questions:

Question 1

The temperature of a bomb calorimeter, and its contents, increased by 8.64 ºC when a 4.127 g sample of ethyne, C_2H_2, was ignited in the bomb. The calibration factor for the apparatus was 23.9 kJ $^{o}C^{-1}$. The thermochemical equation that would best describe this reaction would be:

a) $C_2H_{2(g)} + \frac{5}{2}O_{2(g)} \longrightarrow 2CO_{2(g)} + H_2O_{(l)}$ $\Delta H = -650$ kJ mol^{-1}

b) $C_2H_{2(g)} + \frac{5}{2}O_{2(g)} \longrightarrow 2CO_{2(g)} + H_2O_{(l)}$ $\Delta H = -2600$ kJ mol^{-1}

c) $2C_2H_{2(g)} + 5O_{2(g)} \longrightarrow 4CO_{2(g)} + 2H_2O_{(l)}$ $\Delta H = -1300$ kJ mol^{-1}

d) $2C_2H_{2(g)} + 5O_{2(g)} \longrightarrow 4CO_{2(g)} + 2H_2O_{(l)}$ $\Delta H = -2600$ kJ mol^{-1}

Question 2

Octane, a major component of petrol, burns in air according to the following thermochemical equation.

$$2C_8H_{18(l)} + 25O_{2(g)} \longrightarrow 16CO_{2(g)} + 18H_2O_{(g)} \quad \Delta H = -10,108 \text{ kJ mol}^{-1}$$

Combustion of octane may also be represented by the following thermochemical equation.

$$2C_8H_{18(l)} + 25O_{2(g)} \longrightarrow 16CO_{2(g)} + 18H_2O_{(1)} \quad \Delta H = -10,900 \text{ kJ mol}^{-1}$$

Using the information provided for octane combustion, the enthalphy change (ΔH) for the reaction:

$H_2O_{(l)} \longrightarrow H_2O_{(g)}$ is equal to:

a) + 44 kJ mol^{-1} b) -44 kJ mol^{-1} c) + 792 kJ mol^{-1} d) - 792 kJ mol^{-1}

Question 3

Compared to energy produced from fossil fuels, nuclear energy

a) Is a renewable resource

b) Will produce more energy from a given mass of fuel;

c) Will produce less energy from a given mass of fuel;

d) Is cheaper to set up and operate, making it cheaper to the consumer

Question 4

Hydrogen can be produced commercially using sulfur dioxide and water as starting materials. The overall equation is:

$$H_2O_{(g)} \longrightarrow H_{2\,(g)} + \frac{1}{2}O_{2\,(g)}$$

Using the equations below, determine the enthalpy change for this reaction

$SO_{2(g)} + 2H_2O_{(g)} + Br_{2(g)} \longrightarrow H_2SO_{4(l)} + 2HBr_{(g)}$	$\Delta H = -137.3$ kJ mol^{-1}
$2H_2O_{(g)} + 2SO_{2(g)} + O_{2(g)} \longrightarrow 2H_2SO_{4(l)}$	$\Delta H = -550.8$ kJ mol^{-1}
$2HBr_{(g)} \longrightarrow H_{2(g)} + Br_{2(g)}$	$\Delta H = +103.7$ kJ mol^{-1}

a) -584.4 kJ mol^{-1}	b) + 517.2 kJ mol^{-1}	c) -309 kJ mol^{-1}	d) + 241.8 kJ mol^{-1}

Question 5

Which of the following are factors in determining whether a chemical reaction is endothermic or exothermic?

- **i** **The magnitude of the activation energy**
- **ii** **The strength of the bonds in the reactants and products**
- **iii** **The presence of a catalyst**
- **iv** **The physical states of the reactants**

a) i and ii	b) i, ii and iv	c) all of the above	d) ii and iv

Redox, Galvanic cell, Fuel Cell & Batteries PAST SAC questions:

Question 6

The equations for two spontaneous reactions are:

$$3Cr_{(s)} + 2Ga^{3+}_{(aq)} \longrightarrow 3Cr^{3+}_{(aq)} + 2Ga_{(s)}$$

$$Ga_{(s)} + 3Cu^{2+}_{(aq)} \longrightarrow 3Cu^{+}_{(aq)} + Ga^{3+}_{(aq)}$$

The strongest oxidant according to these equations is:

a) Ga^{3+}	b) Cu^{2+}	c) Cu^{+}	d) Cr^{3+}

Question 7

Which of the following statements is consistent with the observations made during **experiment 1**?

a) The NO_3^- ion is a stronger reductant than metal X .

b) Metal Y is a strong reductant than metal X.

c) The Y^{2+} ion is a stronger oxidant than the X^{2+} ion

d) Water is a stronger oxidant than the X^{2+} ion.

Question 8

In **experiment 2**, which of the following correctly identifies the electrodes in the galvanic cell?

a) Electrode X is the positively charged anode.

b) Electrode X is the negatively charged anode.

c) Electrode Y is the positively charged anode.

d) Electrode Y is the negatively charged anode.

Question 9

Which of the following correctly identifies the flow of charged particles in this cell?

	Flow of cations from the salt bridge to the	**Flow of electrons**
a)	Y^{2+}/Y half-cell	From X to Y
b)	Y^{2+}/Y half-cell	From Y to X
c)	X^{2+}/X half-cell	From X to Y
d)	X^{2+}/X half-cell	From Y to X

Question 10

A fuel cell may be constructed using methanol and oxygen contained in an alkaline electrolyte. The overall reaction for the methanol/oxygen fuel cell is:

$$2CH_3OH_{(l)} + 3O_{2\,(g)} \longrightarrow 2CO_{2\,(g)} + 4H_2O_{(l)}$$

The reaction occurring at the anode would be:

a) $O_2\,(g) + 4H^+_{(aq)} + 4e^- \longrightarrow 2H_2O_{(l)}$

b) $O_2\,(g) + 2H_2O_{(l)} + 4e^- \longrightarrow 4OH^-_{(aq)}$

c) $CH_3OH_{(l)} + H_2O_{(l)} \longrightarrow CO_{2\,(g)} + 6H^+_{(aq)} + 6e^-$

d) $CH_3OH_{(l)} + 6OH^-_{(aq)} \longrightarrow CO_{2(g)} + 5H_2O_{(l)} + 6e^-$

Section B: Short Answer Questions

Answers: Thermochemistry and Energy PAST SAC questions:

Question 1

A strategy to counter Carbon dioxide emissions and reduce the greenhouse effect is to use fuels that provide a better CO_2 efficiency. This efficiency can be calculated as shown below:

$$CO_2 \text{ efficiency} = \frac{\textit{energy released in kJ per kg of fuel}}{\textit{Carbon dioxide emission in grams per kg of fuel}}$$

Consider the fuels, methane CH_4 and ethanol CH_3CH_2OH.

a) Calculate the energy released in kJ per kg of

(i) Methane

$M(CH_4) = 16.0 \text{ g mol}^{-1}$

889 kJ of energy is released by 16.0 g of methane

$\frac{889}{16} = 55.6\ kJ$ **is released by1.0 g of methane**

5.56×10^4 kJ of energy is released per kg of methane [1 mark]

(ii) Ethanol

$M(C_2H_5OH) = 46.0 \text{ g mol}^{-1}$

1364 kJ is released by 46.0 g of ethanol

$\frac{1364}{46} = 29.7\ kJ$ **is released by 1.0 g of ethanol**

2.97×10^4 kJ is released by 1.0 kg of ethanol [1 mark]

b) Calculate the mass, in g, of CO_2 emitted per kg of

(i) Methane

$n(CH_4) = \frac{10^3 g}{16} = 62.5\ mol$ **[1 mark]**

$n(CO_2) = n(CH_4) = 62.5$ mol

$m(CO_2) = n \times M = 62.5 \times 44 = 2750 \text{ g} = 2.75 \times 10^3 \text{g}$ **[1 mark]**

(ii) Ethanol

$n(C_2H_5OH) = \frac{10^3 g}{46} = 21.74\ mol$ [1 mark]

$n(CO_2) = 2n(C_2H_5OH) = 2 \times 21.74 = 43.48$ mol

$m(CO_2) = n \times M = 43.48 \times 44.0 = 1913$ g $= 1.91 \times 10^3$g [1 mark]

c) Calculate the CO_2 efficiency for the two fuels and state which would be more efficient in reducing greenhouse gases.

CO_2 efficiency (methane) = $\frac{5.56 \times 10^4}{1.91 \times 10^3} = 20.2$ [1 mark]

CO_2 efficiency (ethanol) = $\frac{2.97 \times 10^4}{1.91 \times 10^3} = 15.6$ [1 mark]

Methane is more efficient [1 mark]

d) Other than the cost or whether ethanol and methane is derived from a renewable or non-renewable energy source, state another factor that would be considered when selecting between the suitability of ethanol and methane as a fuel

[1 mark] for

- Source of fuel e.g methane is obtained from gas wells whereas ethanol is obtained from sugar cane.
- Storage of fuel
- Availability of fuel to meet the transport/household uses of mankind.

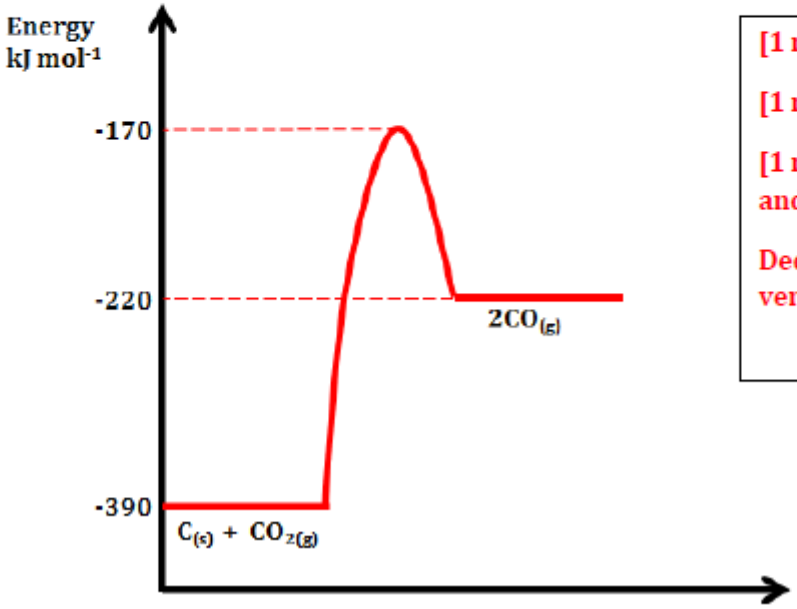
Total 10 marks

Question 2
For the reaction,

$$C_{(s)} + CO_{2\,(g)} \longrightarrow 2CO_{(g)}$$

- The reactants have an enthalpy of – 390 kJ mol^{-1}.
- The forward E_a is 220 kJ mol^{-1}
- The reverse E_a is 22.7% less than the forward E_a

a) Use the above information to draw the energy profile for this reaction.



[1 mark] for all correct values

[1 mark] for correct shape and

[1 mark] labelling the reactants and products and vertical axis

Deduct a mark for no units on vertical axis.

b) How does the stability of the reactants compare to the stability of the products?
The reactants are more stable than the products. [1 mark]

c) What is the ΔH value for this reaction?
ΔH = -220 + 390 = 170 kJ/mol [1 mark for number and units]

d) Calculate the activation energy for the reaction, $CO \longrightarrow \frac{1}{2}CO_{(g)} + \frac{1}{2}C_{(s)}$
The reaction has been halved so and reversed. Therefore the same must be done to the enthalpy value.
$\Delta H = \frac{-170}{2} = -85\ kJ/mol^{-1}$ [1 mark for number and units]

Total 6 marks

Question 3
Biodiesel can be used to run diesel engine motor vehicles and can be used alone or blended with petrol diesel. Biodiesel can be formed from recycled cooking oil in a two stage process.

a. In the first stage the cooking oil is filtered then hydrolysed. What is the purpose of the hydrolysis reaction?

The hydrolysis reaction breaks the three ester links in the triglyceride to produce

the fatty acids. (1 mark)

1 mark

b. During the second stage the biodiesel molecule is produced. Name the type of reaction that is used to produce this molecule and the by-product that is formed.

Reaction: **Esterification or Condensation** (1 mark)

By-product: **H_2O or water** (1 mark)

2 marks

c. Write an overall balanced equation for the production of the biodiesel 'methyl palmitoleate' from its fatty acid (Note: states are not required).

$$C_{15}H_{29}COOH + CH_3OH \xrightarrow{(NaOH/Heat)} C_{15}H_{29}COOCH_3 + H_2O$$

1 mark

d. Explain why Biodiesel can be referred to as a "carbon neutral" fuel source.

Vegetable oil comes from a plant source that consumes CO_2 while the fuel releases CO_2 into the atmosphere (1 mark)

The CO_2 consumed cancels out the CO_2 produced making it carbon neutral (1 mark)

2 marks

Total 6 marks

Answers: Redox, Galvanic cell, Fuel Cell & Batteries PAST SAC questions

Question 4

a) Write the half-equation for the reaction occurring at the cathode when the battery is discharging.

$Br_{2(l)} + 2e \longrightarrow 2Br^-_{(aq)}$ **[1 mark for balanced equation, 1 mark for all correct states]**

b) In what way does the anode change as the battery is discharging?

The anode decreases in mass (because the *zinc* metal is oxidised at the anode during discharge). [1 mark]

c) An important feature of the battery is the special membrane.

i) What role does the membrane have in the operation of the battery during discharge?

The membrane allows for the flow of ions between the electrolyte in the two half-cell Chambers. [1 mark]

ii) Bromine cannot pass through the membrane.

Explain the importance of this restriction when the battery is being recharged.

During recharge, bromine and zinc are produced and must be kept isolated from each other to prevent a spontaneous redox reaction. [1 mark]

c) The zinc-bromine flow battery produces a voltage of 1.67 volts per cell. This is below the predicted value from the electrochemical series. Suggest one possible reason for this lower than expected value.

The conditions used in the cell differ from the standard conditions under which the Electrochemical series is constructed. [1 mark]

d) A useful feature of the battery is its ability to be recharged using renewable energy sources. Outline how the battery could be recharged using solar energy.

Using a photovoltaic (or solar) cell, the Sun's energy can be converted directly to electrical energy. [1 mark]

The solar cell could be put in place of the power source in the diagram, allowing the zinc-bromine battery to be recharged. [1 mark]

Total 8 marks